

François Leroy

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📄 <https://github.com/FrsLry>

Macroecology, Statistical Modelling (ML, DL)

Experience

2024-Ongoing **Postdoctoral researcher - Using Artificial Intelligence to understand spatio-temporal changes of biodiversity**, *The Ohio State University*, dept. of Evolution, Ecology and Organismal Biology, Columbus, Ohio.



- Creating neural hierarchical models from Bayesian framework
- Tailoring loss functions to infer distribution parameters
- Creating simulated data to test models
- Hired for the project [ABC Global Center](#), collaborating with computer scientists from MIT, McGill University, Mila institute (Montreal, Canada)

2020-2024 **PhD. - Spatial scaling and decomposition of macroecological changes**, *Faculty of Environmental Sciences, CZU*, dept. of *Spatial sciences*, Prague.



- Using big ecological data to model biodiversity changes across spatial scales
- Using model optimisation and selection on various machine learning algorithms (Random Forest, BRT, XGBoost, linear models...)

Education

2021 (1 semester) **Machine Learning**, *Faculty of Mathematics and Physics, UFAL, Charles University*, Prague.
Studying all Machine Learning algorithms, from Support Vector Machines to Neural Networks

2020-2024 **PhD. - Spatial scaling and decomposition of macroecological changes**, Prague.



2018-2020 **Marine Sciences MSc**, *Sorbonne University*, Paris.

Numerical Ecology, modelling, geostatistics, GIS, oceanography, marine ecology, biogeochemistry, database management

2015-2018 **Bachelor of Science**, *South Brittany University*, Vannes (France).
Specialized in Coastal Ecosystems and Management, GIS

Teaching

- 2021-2024
- Statistical ecology and macroecology
 - Version Control using Git and Github
 - GIS and spatial analysis

Modelling skills

- Deep Learning** Multilayer Perceptron, Convolutional/Recurrent/Hierarchical Neural Networks
- Machine Learning** Classification and Regression Tree based algorithms (RF, BRT, GBM, XGBoost), Linear Models (GLM, Mixed Models, polynomial regressions...), Hierarchical modelling
- Others** Generalized Additive Models, Bayesian inference with MCMC algorithms, Bayesian Networks, Hidden Markov Models, Time series analysis, Model: optimisation (e.g. regularization), prediction, scalability

Programming skills

- Advanced Python, R, Git, QGIS, ArcGIS, LaTeX
- Intermediate Shell, MySQL
- Basic Julia, MATLAB, HTML5, CSS

Highlighted Internships and others

2024 (1 semester) **Deep Learning**, *Faculty of Mathematics and Physics*, Charles University, Prague.
○ Going through all Deep Learning algorithms

2022 (1 week) **HMSC**, *Jyväskylä summer school*, Jyväskylä, Finland.
○ Summer school on Hierarchical Modeling of Species Community

- 2020 **Community Modelling**, *DYNECO-LEBCO*, *IFREMER*, Brest (France).
(6 months)
 - Bayesian networks and qualitative modelling to assess the impact of environmental changes on benthic communities

Highlighted Talks and Conferences

- Invited speaker 2024-02-16 **Introduction to Reproducible Science: Version Control using Git and Github**, *Ecoinformatics IAVS*, Online, [Slides](#).
- Conference 2023-08-10 **Decomposing abundance change to recruitment and loss: analysis of the North-American avifauna**, *Ecological Society of America*, Portland, OR, [Slides](#).
- Conference 2022-06-05 **Untangling biodiversity changes across a continuum of spatial scales**, *International Biogeography Society conference*, Vancouver, BC, [Slides](#).

Scientific Referees

Dr. **Marta Jarzyna**, Ohio State University, ☎+1 (978) 587-5938, jarzyna.1@osu.edu

Dr. **Petr Keil**, Czech University of Life Sciences, ☎+420 224382659, keil@fzp.czu.cz

Dr. **Martin Marzloff**, Ifremer, ☎+332 98224327, Martin.Marzloff@ifremer.fr

Dr. **Vítězslav Moudrý**, Czech University of Life Sciences, ☎+420 224382653, moudry@fzp.czu.cz

Publications

Petr Keil, Adam T. Clark, Vojtěch Barták, and **François Leroy**. Should Regional Species Loss Be Faster or Slower Than Local Loss? It Depends on Density-Dependent Rate of Death. URL: <https://onlinelibrary.wiley.com/doi/10.1002/ece3.71162>, doi:10.1002/ece3.71162.

Vítězslav Moudrý, Kateřina Gdulová, Lukáš Gábor, Eliška Šárovcová, Vojtěch Barták, **François Leroy**, Olga Špatenková, Duccio Rocchini, and Jiří Prošek. Effects of environmental conditions on ICESat-2 terrain and canopy heights retrievals in central european mountains. URL: <https://www.sciencedirect.com/science/article/pii/S0034425722002267>, doi:10.1016/j.rse.2022.113112.

Vítězslav Moudrý, Petr Keil, Anna F Cord [...], **François Leroy** [...], and Petra Šímová. Scale mismatches between predictor and response variables in species distribution modelling: A review of practices for appropriate grain selection. doi:10.1177/03091333231156362.

Dominika Prajzlerová, Vojtěch Barták, Petr Keil, Vítězslav Moudrý [...], **François Leroy** [...], and Petra Šímová. The relationship between remotely-sensed spectral heterogeneity and bird diversity is modulated by landscape type. URL: <https://linkinghub.elsevier.com/retrieve/pii/S1569843224001171>, doi:10.1016/j.jag.2024.103763.

François Leroy. *Spatial scaling and decomposition of macroecological changes*. Doctoral theses, dissertations, Czech University of Life Sciences Prague, Faculty of Environmental Sciences Praha, 2024. SUPERVISOR: Mgr. Petr Keil, Ph.D. URL: <https://theses.cz/id/aigse5/>.

François Leroy, Jiří Reif, David Storch, and Petr Keil. How has bird biodiversity changed over time? a review across spatio-temporal scales. 69:26–38. URL: <https://www.sciencedirect.com/science/article/pii/S1439179123000117>, doi:10.1016/j.baee.2023.03.004.

François Leroy, Jiří Reif, Zdeněk Vermouzek, Karel Šťastný, Eva Trávníčková, Vladimír Bejček, Ivan Mikuláš, and Petr Keil. Decomposing biodiversity change to processes of extinction, colonization, and recurrence across scales. URL: <https://onlinelibrary.wiley.com/doi/10.1111/ecog.06995>, doi:10.1111/ecog.06995.